Sanaul Haque

Rudro_Karmokar_Proposal

Course 608

Document Details

Submission ID
trn:oid:::3117:635610323

Submission Date
Oct 7, 2026, 10:37 PM GMT+6

Download Date
Oct 7, 2026, 10:39 PM GMT+6

File Name
Rudro_Karmokar_Proposal.pdf

File Size
361.5 KB

7 Pages

1,385 Words

7,427 Characters

28% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

- **22** Not Cited or Quoted 22%
  Matches with neither in-text citation nor quotation marks
- **4** Missing Quotations 3%
  Matches that are still very similar to source material
- **4** Missing Citation 4%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

25% Internet sources
12% Publications
6% Submitted works (Student Papers)

Match Groups

- **22** Not Cited or Quoted 22%
  Matches with neither in-text citation nor quotation marks
- **4** Missing Quotations 3%
  Matches that are still very similar to source material
- **4** Missing Citation 4%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

25% Internet sources
12% Publications
6% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Internet	iarco.org 10%
2	Internet	hal.science 6%
3	Publication	A. Yu. Khnykov, A. Yu. Vdovichenko, S. N. Chvalun. "Nanostructured Materials for ... 3%
4	Publication	Durvesh Gautam, Yogendra K. Gautam, Kavita Sharma, Ashwani Kumar, Ajay Ku... 2%
5	Internet	hal-amu.archives-ouvertes.fr 2%
6	Internet	pmc.ncbi.nlm.nih.gov 2%
7	Publication	V. Krivetskiy, A. Ponzoni, E. Comini, S. Badalyan, M. Rumyantseva, A. Gaskov. "Sele... 1%
8	Internet	pubs.rsc.org <1%
9	Internet	www.springerprofessional.de <1%
10	Publication	Koichi Suematsu, Yuka Shin, Zhongqiu Hua, Kohei Yoshida, Masayoshi Yuasa, Tets... <1%

11 Publication

Pravas Kumar Panigrahi, Basavaiah Chandu, Nagaprasad Puvvada. "Recent Adva... <1%

IARCO RESEARCH PROPOSAL

Full Legal Name: Rudro Karmokar

Institution: Rajshahi University of Engineering & Technology

Category: Senior

Class/Grade/Year: BSc

Country: Bangladesh

Major (if senior): Electrical & Electronic Engineering

Submission Date: September 20, 2026

Registered Email Address: abirkarmokar01@gmail.com

Research Topic: Design Optimization & Material Selection of a Gas Chamber: A Finite Element Method Analysis of Geometrical Shapes

Title: **Design Optimization & Material Selection of a Gas Chamber: A Finite Element Method Analysis of Geometrical Shapes**

Research Problem

The chamber used to test gas sensors needs to be properly sized and optimized. In fact, the reaction and recovery times, along with the amplitude, are highly reliant on the testing environment [1]. The use of very effective gas sensors in gas chambers as drug detection is crucial both now and in the future. However, there are certain issues with the way gas chambers are being developed. Gas chambers are currently of relatively low quality and working efficiency, and issues like aging and inappropriate processes are present [2]. Numerous studies have been published thus far on the improvement of gas detecting capabilities, either through alteration of nanoparticles, or by the nano metal oxide-sensitive material structure [3]. But Few studies have examined how the gas testing system effects on the performance of the sensors. Reaching a trustworthy and extremely sensitive sensor that responds and recovers quickly; however, it is unable to be carried out without a well-designed gas testing chamber. It is necessary to model and simulate such a design in terms of gas flow direction, sensor location, and chamber volume [2]. In this proposal, we will try to simulate the finite element method for three shaped chambers named cylindrical shaped, perfect square and rectangular shape. Then we will go through various materials to choose the perfect one for our chamber based on various criteria like cost, safety factors and the weight our chamber will carry.

Existing Literature

So basically, there was some work with two types of chambers before, the shapes were cross one and boat one. The first one had a total volume of 0.3 L and a gas flow rate of 0.1 L/min to 1.5 L/min. It was constructed of stainless steel. Additionally, it has an inlet and an outlet, and the sensor's ideal location is in the middle, ahead of the direction of the gas flow. Then in next shape, its modest volume of 2.35 10^3 L and gas flow range of 0.01 L/min to 0.5 L/min were produced from polylactic acid (PLA). Its inlet and outflow are planar, giving it a boat-like shape. Furthermore, the ideal sensor location is in the middle of the boat's backside, paralleling the direction of gas movement. The gas flow was kept constant at 0.1 L/min in both chambers [1].

A potential method for producing selective sensor materials involves using a catalyst to modify the surface to create certain centers that take part in the reaction with the analyte gas. Because these centers can react with gas molecules by creating surface complexes, the modifier's chemical makeup, oxidation-reduction capabilities, and acid-base characteristics play a critical role in how the sensor reacts to the target molecule. A wide range of noble metals and oxides (Au, PdO, RuO2, NiO, CuO, Fe2O3, La2O3, V2O5, MoO3, and Sb2O5) were used to modify a SnO2-based sensor in [4], and possibilities for introducing different nanoscale catalysts to produce sensor materials with enhanced selectivity were identified [5].

Lin et al. studied a SnO2 nanowire Hall effect gas sensor for hydrogen detection [6]. It produced stainless steel using a horizontal electric furnace to create SnO2 nanowires. steel mesh, and the authors determined its crystal structure, shape, and electron binding energy using XRD and XPS. In the perspective, the detection procedure of the SnO2 nanowire Hall effect gas sensor was also discussed. The synthesized SnO2 NW sensor displays responses to H2 gas at different operating temperatures and H2 concentrations.

Research Question

This study aims to evaluate the efficiency and accuracy of which shape we should consider for our chamber. Additionally, which materials would be perfect choice for the chamber to be made of. In this respect, the following research question has been addressed:

1. What are the key design considerations for developing a high-performance gas chamber for sensor testing in drug detection, and how do chamber shape and material properties contribute to these factors?
2. How can finite element simulations be used to optimize gas chamber design (cylindrical, square, rectangular) for improved sensor performance?
3. What is the impact of different chamber geometries (cylindrical, square, rectangular) on the gas flow dynamics and sensor performance?

Methodology

The method is fully finite element method based. The software which was used is MATLAB.

Firstly, I went through every shape individually in MATLAB using finite element approach. The Pde toolbox was used throughout the analysis of every shape. Then

dimensions of every shape were analyzed to see which will fit our criteria based on Von Moses Stress distribution and the safety factor of each shape. Thoroughly we used different dimensions to get our desired one. Then we compared all the shapes and found out the cylindrical shape would be best one according to our result. As the other two shapes would have dead zone for their four corners.

After that I went to see which material would be perfect for our chamber. We took in all types of environments the chamber would face while gas sensing using the sensor. The thermal expansion, internal and external pressure, the effect of high temperature (up to 500 degrees centigrade) were few major things to analyze for both the cases mentioned above and this one. Then material database and the prices of all materials in Bangladesh market were analyzed. Then we came up with the fact that stainless steel 316L would be best fit for its low cost, durable weight to design the chamber, and obviously the safety factor.

The code can be provided if asked further.

Research Topic

1 **Potential Limitation:** The model is designed to capture complicated situations with insufficient gauge data, hence affecting the accuracy of the overall safety factor and stress.

Conclusion: This simulation data will be used to enhance gas sensing of different gas using desired chamber. This study offers a key instrument for gas sensor and chamber analysis and simulation knowledge which can be further developed.

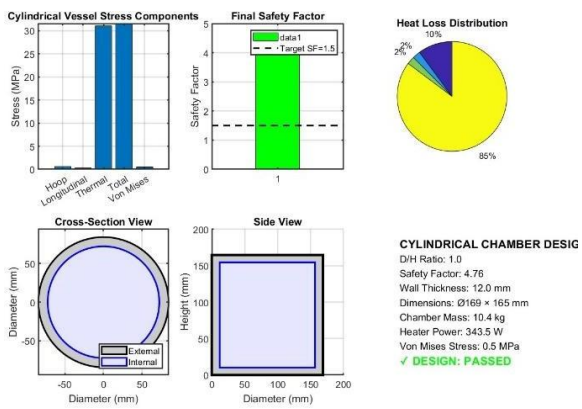
1 **Project Practicalities**: The project will take 3 months only. The first 1 months will be used to collect data and preprocess, which we already did and subsequent data analysis to make the chamber and feed with all the electrical connection would take less than 1 1 month. The last month will be reserved for writing the final report.

Post-program Plan: After further collaboration with researchers for guidance and refinement, the results will be published as a journal paper.

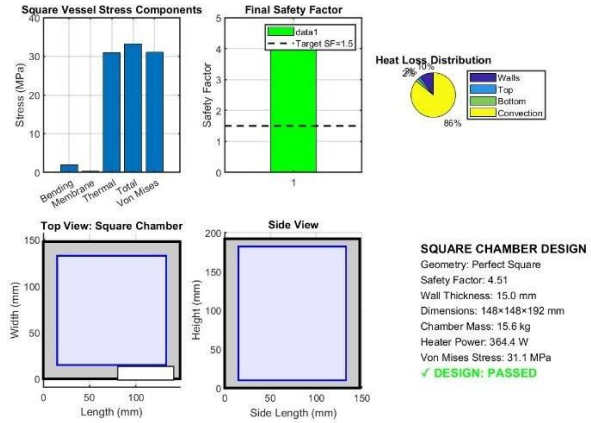
1 **Quality of Writing**

The language of this proposal is academically precise, clear, and appropriate for a 1 multidisciplinary audience. The structure follows standard research norms. The proposal adheres to grammar, citation, and formatting guidelines strictly to meet IARCO standards.

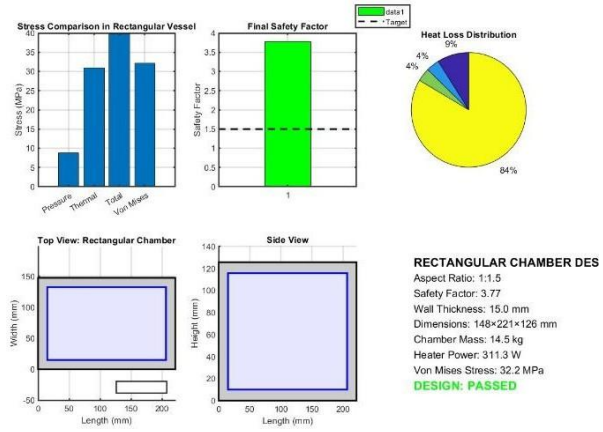
Expected Outcome (Some advancement of my work)



(a)



(b)



(c)

Fig 1: (a) Analysis of Cylindrical shape, (b) Perfect Square, (c) Rectangular Gas Chamber

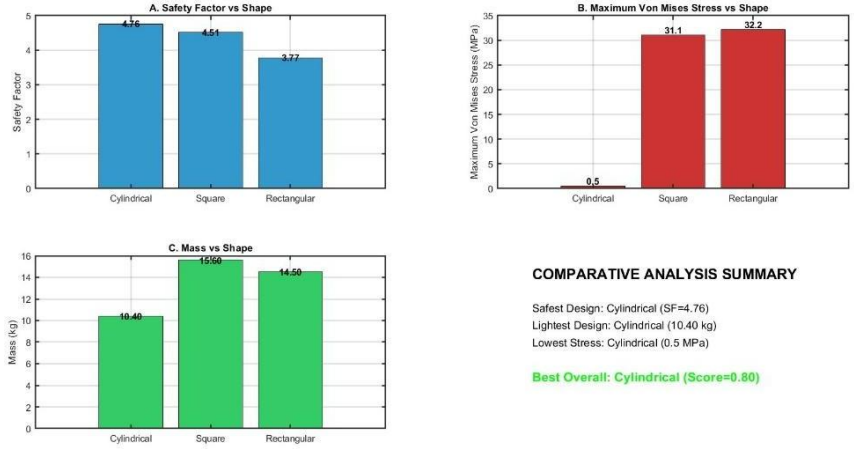


Fig 2: Comparison between three shapes to determine which is better suited for chamber

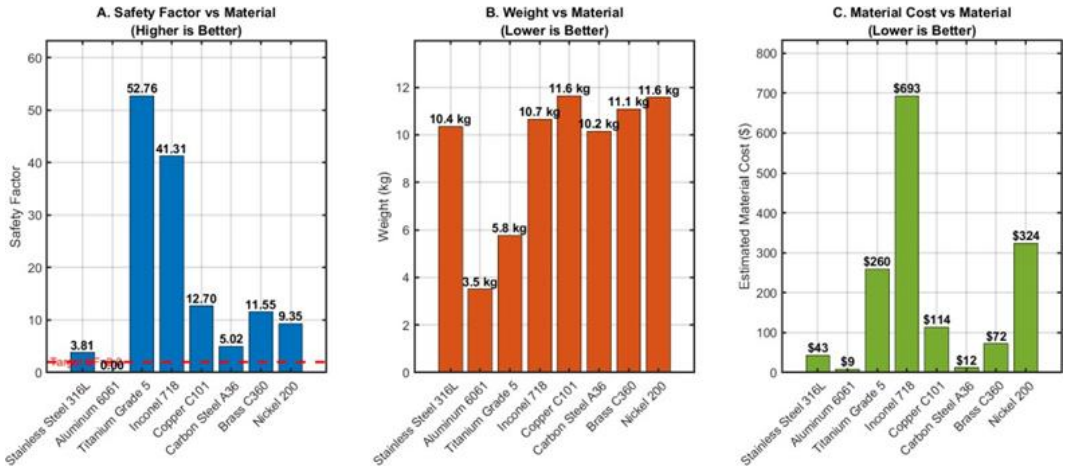


Fig 3: Comparison between different materials to determine which is better suited for chamber

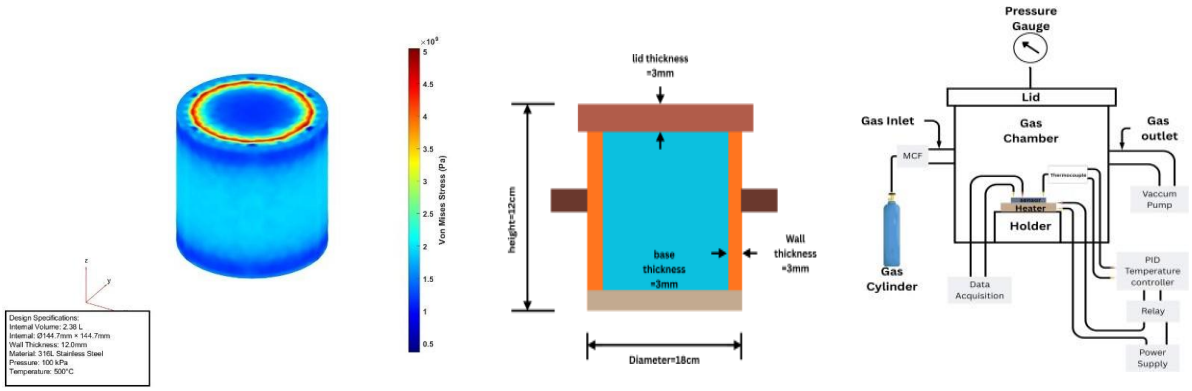


Fig 4: Von Moses Stress analysis and whole electrical connection of the Cylindrical Shape

References

[1] F. E. Annanouch, "Design and optimization of gas sensor testing chamber," 2024. [Online]. Available: https://hal.science/hal-03942498v1

[2] D. L. and K. Xu, "Recent progress in gas sensor based on nanomaterials," MDPI, 2024. [Online]. Available: https://www.mdpi.com/

[3] F. Annanouch et al., "Aerosol-assisted CVD-grown PdO nanoparticle-decorated tungsten oxide nanoneedles extremely sensitive and selective to hydrogen," ACS Appl. Mater. Interfaces, vol. 8, pp. 10413–10421, Apr. 2016.

[4] V. Krivetskiy, A. Ponzoni, E. Comini, et al., "Selectivity modification of SnO2-based gas sensors," Electroanalysis, vol. 22, no. 24, pp. 2809–2816, 2010. doi: 10.1002/elan.201000277

[5] A. Y. Khnykova, "Nanostructured materials for gas sensors," 2024. doi: 10.1134/S2635167625600051

[6] J. Y. Lin, X. L. He, A. J. Zhang, S. H. Huang, and Z. X. Chen, "A Hall effect hydrogen-selective gas sensor based on SnO2 nanowires operating at low temperature," J. Mater. Sci.: Mater. Electron., vol. 30, pp. 20696–20702, 2019.